# Lesson Planning

## Complete Lesson Plan Book

**Class: Class 11 – FSc Pre-Engineering**

**Subject: Chemistry**

**Medium: English**

Academic Year 2026-27

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This book is prepared as a teaching guide for teachers.

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# Unit 1: Basic Concepts

## Learning Objectives

• Define and explain mole concept in the context of Basic Concepts.

• Differentiate between stoichiometry and limiting reactant with suitable examples.

• Solve numerical problems related to empirical formula accurately.

• Apply the ideas of Basic Concepts to everyday situations involving percentage yield.

## Key Concepts

• Meaning, definition and scope of mole concept.

• Important properties, types and examples of stoichiometry.

• Relationship between limiting reactant and empirical formula.

• Practical applications of Basic Concepts, especially percentage yield.

• Key terms and a summary of the main points of Basic Concepts.

## Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Basic Concepts with a daily-life example.
2. Explain mole concept on the board with definitions and clear diagrams.
3. Discuss stoichiometry and limiting reactant using simple, relatable examples.
4. Involve the students through short questions on empirical formula.
5. Solve a numerical on percentage yield together with the class.
6. Summarize the lesson and highlight the key points of Basic Concepts.

## Classroom Activities

• Group discussion: students compare stoichiometry with limiting reactant.

• Board activity: solve a question on empirical formula in pairs.

• Quick quiz on the key terms of Basic Concepts.

## Assessment Questions

1. Define mole concept.
2. Differentiate between stoichiometry and limiting reactant.
3. Solve a numerical based on empirical formula.
4. Write a short note on percentage yield.

## Homework

• Read Basic Concepts from the book and solve the exercise questions.

• Prepare short notes on mole concept and stoichiometry.

## Time Allocation

4–6 periods of 40 minutes each

# Unit 2: Experimental Techniques

## Learning Objectives

• Define and explain filtration in the context of Experimental Techniques.

• Differentiate between crystallization and sublimation with suitable examples.

• Solve numerical problems related to solvent extraction accurately.

• Apply the ideas of Experimental Techniques to everyday situations involving chromatography.

## Key Concepts

• Meaning, definition and scope of filtration.

• Important properties, types and examples of crystallization.

• Relationship between sublimation and solvent extraction.

• Practical applications of Experimental Techniques, especially chromatography.

• Key terms and a summary of the main points of Experimental Techniques.

## Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Experimental Techniques with a daily-life example.
2. Explain filtration on the board with definitions and clear diagrams.
3. Discuss crystallization and sublimation using simple, relatable examples.
4. Involve the students through short questions on solvent extraction.
5. Solve a numerical on chromatography together with the class.
6. Summarize the lesson and highlight the key points of Experimental Techniques.

## Classroom Activities

• Group discussion: students compare crystallization with sublimation.

• Board activity: solve a question on solvent extraction in pairs.

• Quick quiz on the key terms of Experimental Techniques.

## Assessment Questions

1. Define filtration.
2. Differentiate between crystallization and sublimation.
3. Solve a numerical based on solvent extraction.
4. Write a short note on chromatography.

## Homework

• Read Experimental Techniques from the book and solve the exercise questions.

• Prepare short notes on filtration and crystallization.

## Time Allocation

4–6 periods of 40 minutes each

# Unit 3: Gases

## Learning Objectives

• Define and explain gas laws in the context of Gases.

• Differentiate between kinetic molecular theory and diffusion with suitable examples.

• Solve numerical problems related to real gases accurately.

• Apply the ideas of Gases to everyday situations involving Dalton's law.

## Key Concepts

• Meaning, definition and scope of gas laws.

• Important properties, types and examples of kinetic molecular theory.

• Relationship between diffusion and real gases.

• Practical applications of Gases, especially Dalton's law.

• Key terms and a summary of the main points of Gases.

## Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Gases with a daily-life example.
2. Explain gas laws on the board with definitions and clear diagrams.
3. Discuss kinetic molecular theory and diffusion using simple, relatable examples.
4. Involve the students through short questions on real gases.
5. Solve a numerical on Dalton's law together with the class.
6. Summarize the lesson and highlight the key points of Gases.

## Classroom Activities

• Group discussion: students compare kinetic molecular theory with diffusion.

• Board activity: solve a question on real gases in pairs.

• Quick quiz on the key terms of Gases.

## Assessment Questions

1. Define gas laws.
2. Differentiate between kinetic molecular theory and diffusion.
3. Solve a numerical based on real gases.
4. Write a short note on Dalton's law.

## Homework

• Read Gases from the book and solve the exercise questions.

• Prepare short notes on gas laws and kinetic molecular theory.

## Time Allocation

4–6 periods of 40 minutes each

# Unit 4: Liquids and Solids

## Learning Objectives

• Define and explain intermolecular forces in the context of Liquids and Solids.
• Differentiate between vapour pressure and crystal lattices with suitable examples.
• Solve numerical problems related to hydrogen bonding accurately.
• Apply the ideas of Liquids and Solids to everyday situations involving viscosity.

## Key Concepts

• Meaning, definition and scope of intermolecular forces.
• Important properties, types and examples of vapour pressure.
• Relationship between crystal lattices and hydrogen bonding.
• Practical applications of Liquids and Solids, especially viscosity.
• Key terms and a summary of the main points of Liquids and Solids.

## Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Liquids and Solids with a daily-life example.
2. Explain intermolecular forces on the board with definitions and clear diagrams.
3. Discuss vapour pressure and crystal lattices using simple, relatable examples.
4. Involve the students through short questions on hydrogen bonding.
5. Solve a numerical on viscosity together with the class.
6. Summarize the lesson and highlight the key points of Liquids and Solids.

## Classroom Activities

• Group discussion: students compare vapour pressure with crystal lattices.
• Board activity: solve a question on hydrogen bonding in pairs.
• Quick quiz on the key terms of Liquids and Solids.

## Assessment Questions

1. Define intermolecular forces.
2. Differentiate between vapour pressure and crystal lattices.
3. Solve a numerical based on hydrogen bonding.
4. Write a short note on viscosity.

## Homework

• Read Liquids and Solids from the book and solve the exercise questions.
• Prepare short notes on intermolecular forces and vapour pressure.

## Time Allocation

4–6 periods of 40 minutes each

# Unit 5: Atomic Structure

## Learning Objectives

• Define and explain discovery of electron in the context of Atomic Structure.

• Differentiate between Bohr's model and quantum numbers with suitable examples.

• Solve numerical problems related to orbitals accurately.

• Apply the ideas of Atomic Structure to everyday situations involving electronic configuration.

## Key Concepts

• Meaning, definition and scope of discovery of electron.

• Important properties, types and examples of Bohr's model.

• Relationship between quantum numbers and orbitals.

• Practical applications of Atomic Structure, especially electronic configuration.

• Key terms and a summary of the main points of Atomic Structure.

## Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Atomic Structure with a daily-life example.
2. Explain discovery of electron on the board with definitions and clear diagrams.
3. Discuss Bohr's model and quantum numbers using simple, relatable examples.
4. Involve the students through short questions on orbitals.
5. Solve a numerical on electronic configuration together with the class.
6. Summarize the lesson and highlight the key points of Atomic Structure.

## Classroom Activities

• Group discussion: students compare Bohr's model with quantum numbers.

• Board activity: solve a question on orbitals in pairs.

• Quick quiz on the key terms of Atomic Structure.

## Assessment Questions

1. Define discovery of electron.
2. Differentiate between Bohr's model and quantum numbers.
3. Solve a numerical based on orbitals.
4. Write a short note on electronic configuration.

## Homework

• Read Atomic Structure from the book and solve the exercise questions.

• Prepare short notes on discovery of electron and Bohr's model.

## Time Allocation

4–6 periods of 40 minutes each

# Unit 6: Chemical Bonding

## Learning Objectives

• Define and explain ionic bond in the context of Chemical Bonding.

• Differentiate between covalent bond and VSEPR theory with suitable examples.

• Solve numerical problems related to hybridization accurately.

• Apply the ideas of Chemical Bonding to everyday situations involving metallic bond.

## Key Concepts

• Meaning, definition and scope of ionic bond.

• Important properties, types and examples of covalent bond.

• Relationship between VSEPR theory and hybridization.

• Practical applications of Chemical Bonding, especially metallic bond.

• Key terms and a summary of the main points of Chemical Bonding.

## Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Chemical Bonding with a daily-life example.
2. Explain ionic bond on the board with definitions and clear diagrams.
3. Discuss covalent bond and VSEPR theory using simple, relatable examples.
4. Involve the students through short questions on hybridization.
5. Solve a numerical on metallic bond together with the class.
6. Summarize the lesson and highlight the key points of Chemical Bonding.

## Classroom Activities

• Group discussion: students compare covalent bond with VSEPR theory.

• Board activity: solve a question on hybridization in pairs.

• Quick quiz on the key terms of Chemical Bonding.

## Assessment Questions

1. Define ionic bond.
2. Differentiate between covalent bond and VSEPR theory.
3. Solve a numerical based on hybridization.
4. Write a short note on metallic bond.

## Homework

• Read Chemical Bonding from the book and solve the exercise questions.

• Prepare short notes on ionic bond and covalent bond.

## Time Allocation

4–6 periods of 40 minutes each

# Unit 7: Thermochemistry

## Learning Objectives

• Define and explain enthalpy in the context of Thermochemistry.
• Differentiate between Hess's law and calorimetry with suitable examples.
• Solve numerical problems related to exothermic reactions accurately.
• Apply the ideas of Thermochemistry to everyday situations involving bond energies.

## Key Concepts

• Meaning, definition and scope of enthalpy.
• Important properties, types and examples of Hess's law.
• Relationship between calorimetry and exothermic reactions.
• Practical applications of Thermochemistry, especially bond energies.
• Key terms and a summary of the main points of Thermochemistry.

## Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Thermochemistry with a daily-life example.
2. Explain enthalpy on the board with definitions and clear diagrams.
3. Discuss Hess's law and calorimetry using simple, relatable examples.
4. Involve the students through short questions on exothermic reactions.
5. Solve a numerical on bond energies together with the class.
6. Summarize the lesson and highlight the key points of Thermochemistry.

## Classroom Activities

• Group discussion: students compare Hess's law with calorimetry.
• Board activity: solve a question on exothermic reactions in pairs.
• Quick quiz on the key terms of Thermochemistry.

## Assessment Questions

1. Define enthalpy.
2. Differentiate between Hess's law and calorimetry.
3. Solve a numerical based on exothermic reactions.
4. Write a short note on bond energies.

## Homework

• Read Thermochemistry from the book and solve the exercise questions.
• Prepare short notes on enthalpy and Hess's law.

## Time Allocation

4–6 periods of 40 minutes each

# Unit 8: Chemical Equilibrium

## Learning Objectives

• Define and explain law of mass action in the context of Chemical Equilibrium.

• Differentiate between equilibrium constant and Le Chatelier's principle with suitable examples.

• Solve numerical problems related to ionic equilibrium accurately.

• Apply the ideas of Chemical Equilibrium to everyday situations involving solubility product.

## Key Concepts

• Meaning, definition and scope of law of mass action.

• Important properties, types and examples of equilibrium constant.

• Relationship between Le Chatelier's principle and ionic equilibrium.

• Practical applications of Chemical Equilibrium, especially solubility product.

• Key terms and a summary of the main points of Chemical Equilibrium.

## Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Chemical Equilibrium with a daily-life example.
2. Explain law of mass action on the board with definitions and clear diagrams.
3. Discuss equilibrium constant and Le Chatelier's principle using simple, relatable examples.
4. Involve the students through short questions on ionic equilibrium.
5. Solve a numerical on solubility product together with the class.
6. Summarize the lesson and highlight the key points of Chemical Equilibrium.

## Classroom Activities

• Group discussion: students compare equilibrium constant with Le Chatelier's principle.

• Board activity: solve a question on ionic equilibrium in pairs.

• Quick quiz on the key terms of Chemical Equilibrium.

## Assessment Questions

1. Define law of mass action.
2. Differentiate between equilibrium constant and Le Chatelier's principle.
3. Solve a numerical based on ionic equilibrium.
4. Write a short note on solubility product.

## Homework

• Read Chemical Equilibrium from the book and solve the exercise questions.

• Prepare short notes on law of mass action and equilibrium constant.

## Time Allocation

4–6 periods of 40 minutes each

# Unit 9: Solutions

## Learning Objectives

• Define and explain concentration units in the context of Solutions.
• Differentiate between Raoult's law and colligative properties with suitable examples.
• Solve numerical problems related to ideal solutions accurately.
• Apply the ideas of Solutions to everyday situations involving Henry's law.

## Key Concepts

• Meaning, definition and scope of concentration units.
• Important properties, types and examples of Raoult's law.
• Relationship between colligative properties and ideal solutions.
• Practical applications of Solutions, especially Henry's law.
• Key terms and a summary of the main points of Solutions.

## Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Solutions with a daily-life example.
2. Explain concentration units on the board with definitions and clear diagrams.
3. Discuss Raoult's law and colligative properties using simple, relatable examples.
4. Involve the students through short questions on ideal solutions.
5. Solve a numerical on Henry's law together with the class.
6. Summarize the lesson and highlight the key points of Solutions.

## Classroom Activities

• Group discussion: students compare Raoult's law with colligative properties.
• Board activity: solve a question on ideal solutions in pairs.
• Quick quiz on the key terms of Solutions.

## Assessment Questions

1. Define concentration units.
2. Differentiate between Raoult's law and colligative properties.
3. Solve a numerical based on ideal solutions.
4. Write a short note on Henry's law.

## Homework

• Read Solutions from the book and solve the exercise questions.
• Prepare short notes on concentration units and Raoult's law.

## Time Allocation

4–6 periods of 40 minutes each

# Unit 10: Electrochemistry

## Learning Objectives

• Define and explain oxidation and reduction in the context of Electrochemistry.

• Differentiate between electrochemical cells and standard electrode potential with suitable examples.

• Solve numerical problems related to electrolysis accurately.

• Apply the ideas of Electrochemistry to everyday situations involving corrosion.

## Key Concepts

• Meaning, definition and scope of oxidation and reduction.

• Important properties, types and examples of electrochemical cells.

• Relationship between standard electrode potential and electrolysis.

• Practical applications of Electrochemistry, especially corrosion.

• Key terms and a summary of the main points of Electrochemistry.

## Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Electrochemistry with a daily-life example.
2. Explain oxidation and reduction on the board with definitions and clear diagrams.
3. Discuss electrochemical cells and standard electrode potential using simple, relatable examples.
4. Involve the students through short questions on electrolysis.
5. Solve a numerical on corrosion together with the class.
6. Summarize the lesson and highlight the key points of Electrochemistry.

## Classroom Activities

• Group discussion: students compare electrochemical cells with standard electrode potential.

• Board activity: solve a question on electrolysis in pairs.

• Quick quiz on the key terms of Electrochemistry.

## Assessment Questions

1. Define oxidation and reduction.
2. Differentiate between electrochemical cells and standard electrode potential.
3. Solve a numerical based on electrolysis.
4. Write a short note on corrosion.

## Homework

• Read Electrochemistry from the book and solve the exercise questions.

• Prepare short notes on oxidation and reduction and electrochemical cells.

## Time Allocation

4–6 periods of 40 minutes each